

Introduction to refrigeration system (Chapter :9) Thermal science

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Topics of discussion

- Refrigeration
- Refrigerator
- Refrigerants
- Refrigeration cycle and its types
- Components of refrigeration cycle

Refrigeration, refrigerator and refrigerants

Refrigeration

Refrigeration may be defined as a process of removing heat from a substance and pumping it to the surroundings. It also includes the process of maintaining and reducing the temperature of a body below the general temperature of its surroundings.

Refrigerator

The equipment used for removing the heat continuously for maintaining a low temperature in a space is called 'refrigerator'.

Refrigerants

The working fluids used for carrying away heat are called 'refrigerants'. Example: Ammonia, SO₂, CO₂, R-134a etc.

Refrigeration cycle

Refrigeration cycle

Equipment used for removing the heat continuously for maintaining a low temperature in a space is called 'refrigerator' and the Cycle on which it operates is called the refrigeration cycle

Types of refrigeration

There are various types of Refrigeration Cycles but two types of Refrigeration are important and are as follows

- 1.Vapor Absorption Refrigeration Cycle
- 2.Vapor Compression Refrigeration Cycle

Refrigeration cycle contd...

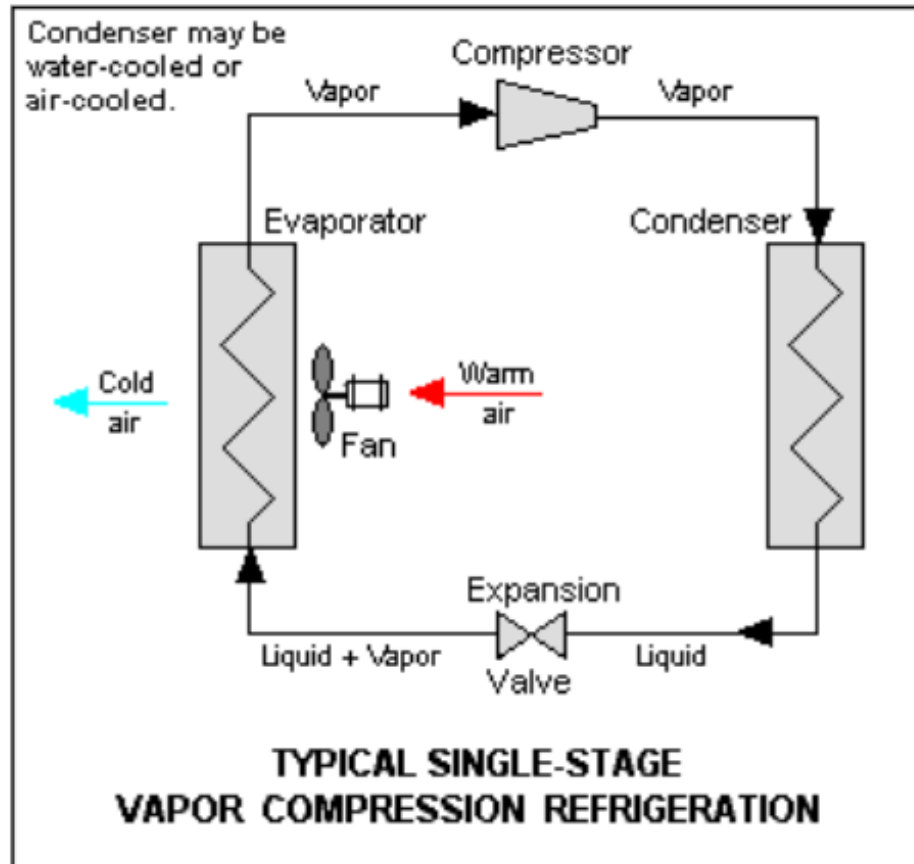
Vapor Compression Refrigeration Cycle (VCRS)

Most important system from the point of commercial and domestic utility and most practical form of refrigeration

Components of the Vapor Compression Refrigeration cycle

- 1.Compressor,
- 2.Condenser,
- 3.Expansion Valve, and
- 4.Evaporator.

Refrigeration cycle contd...

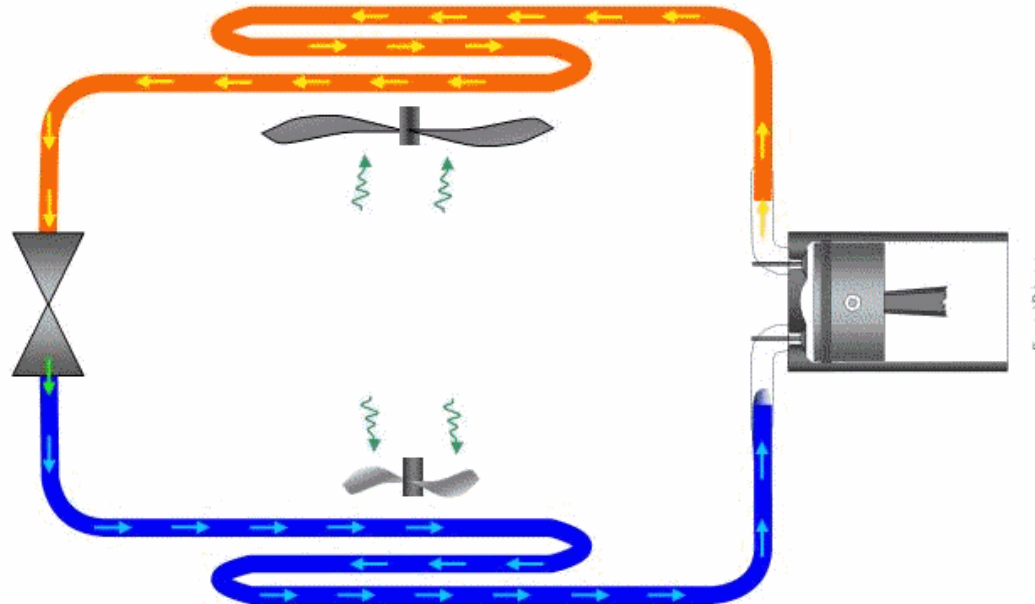


Refrigeration cycle contd...

The working of Vapor Compression

Refrigeration System can be completed under 4 processes which are as

- 1.Compression,
- 2.Condensation,
- 3.Expansion and
- 4.Vaporization process.



THANK YOU

Stay home



Stay safe